Multi Agent Systems

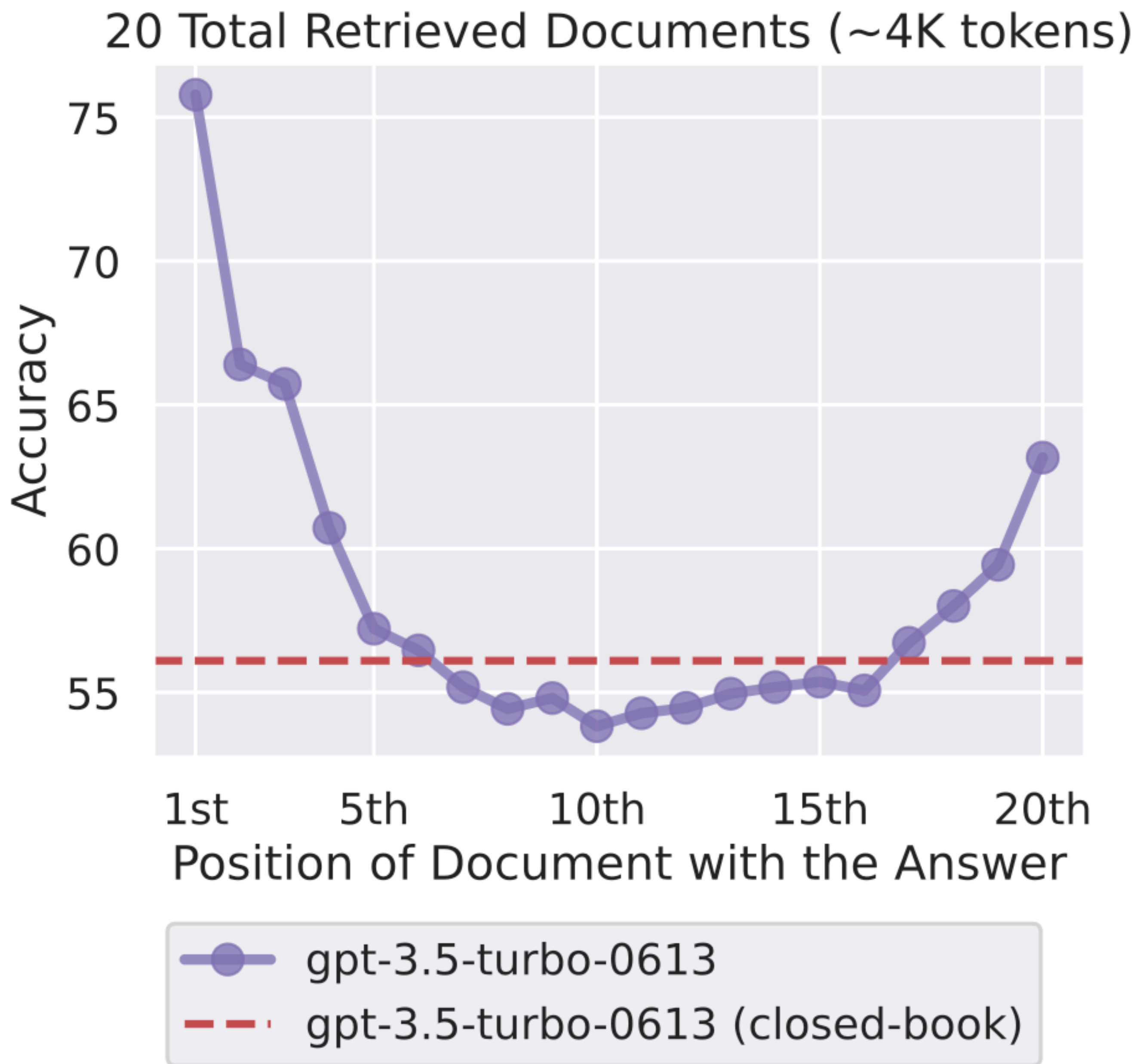
Protocols Over Personas

Chatbots

- Agents are wrappers around chatbots, that have access to tools, memory, etc.
- Why would one Agent not suffice?
    - If the task was complex… why not just give it a long and complex prompt?
    - Or a sequence of prompts that tell the agent to accomplish each step?

A single chat-based LLM session is

- Finite context windows
- Is stateless and pushes transcript-and-state management onto the developer
- Token Snow Ball - Has quadratic rising cost/latency when you naively “stuff the prompt”
- Is a black-box, entirely
- Persona Contradiction
- Is brittle and does not pay equal attention, when a monolithic prompt merges all of these into one prompt persona.
    - planning
    - execution
    - verification
    - safety
    - UI interaction



Source: Liu et. al. “Lost in the Middle: How Language Models Use Long Contexts” 4

Multi-Agent

- does not simply mean **<u>“more intelligent”</u>**
- is an engineering approach to improve
  - modularity
  - parallelism
  - reliability
  - governance
  - evaluability
- while introducing coordination overhead, new failure modes, security risks, and higher cost if misapplied.

Vibe-Based Multi-Agent

- Put 100 extremely smart people in a room.
  - Tell them they are all equal.
  - Give them one instruction:
  - ‘Figure out how to land a human on the moon.’ Then leave.”
- What happens?
- Did intelligence fail? Or did coordination fail?

Structured Multi-Agent

- Same 100 smart people.
- But this time: Define roles (propulsion, life support, materials, trajectory, risk). Appoint a mission director
    - Define decision rights
    - Establish proposal formats
    - Set review gates
    - Enforce budget constraints
    - Define termination criteria
    - Require written artifacts
    - Require safety sign-off
- Now you don’t just have smart people.
    - You have NASA.
- That’s protocol.

Core Principles

- Protocol beats Vibe
- Coordination over intelligence
- Separation of Objectives (and guardrails)
- Parallelism
- Reliability through redundancy - Ensembles, debate, and cross-checking can reduce errors
    - independent attempts + aggregation frequently beat a single attempt.
- Governance and security boundaries (HITL)
- Evaluability and debugging

Protocol beats Vibe

- Debate: Should advanced AI systems require FDA-style pre-deployment approval?
  - Vibe
  - Structured debate
- Intelligence without structure becomes noise at scale.
- Large multi-agent systems fail not because models are weak but because coordination is underspecified.
- “Agents” are components and protocols are the architecture.

Coordination over raw intelligence

- Overall system's capability comes from coordination, protocols, and evaluation, not from any single "perfect" prompt.
  - who speaks when, who decides, who can veto, how conflicts resolve, and how partial results merge.
- An orchestrator agent does not attempt to read a 100-page document.
- Instead, it delegates the task to multiple parallel subagents, each provided with a narrowly defined chunk of the text and a highly specific objective.

Common Architectures

- Orchestrator-Worker (Supervisor)
- Hierarchical
- Blackboard patterns
- Market-Based and Negotiation Protocols
- Graph based

- To implement these topologies, the engineering ecosystem has shifted from handcrafted logic scripts to robust, framework-driven development.

Frameworks

- Conversational multi-agent systems
    - AutoGen (conversational control)
    - Swarm (conversational control)
    - CrewAI (largely sequential)

- Structured control systems
    - LangGraph (state machine)
    - Semantic Kernel SK (event-driven steps)

MCP vs A2A

- MCP - how models/LLMs talk to tools
- Agent-to-Agent Protocol (A2A)
  - Standard for communication between multiple agents
  - Enables Collaboration
  - Systems span multiple services or organizations

Emergent Behavior

- **Emergence** occurs when the collective output of a multi-agent system exhibits complex, coordinated intelligence and behavioral phenomena that were never explicitly programmed into any single agent's foundational code.

- Stanford Smallville - The system demonstrated that bounded AI agents, when interacting through robust protocols, could successfully model complex, long-term human social dynamics.

- Multi-agent reinforcement learning (MARL) - Without being hardcoded with economic theories, the agents learned the mechanics of production, consumption, and the trading of physical goods.

Source: Park JS et. al. “Generative Agents: Interactive Simulacra of Human Behavior”